

UX CASE STUDY

Clinic Appointment Booking

Redesigning Healthcare Access in Senegal

From missed calls and long waits to a simple, reassuring digital booking experience.

Project Overview

Context - Goal - Business Objective

The Mission

Patients in local clinics across Senegal struggle with long wait times and an unreliable, frustrating appointment booking process. Clinics rely on phone calls that often go unanswered. Patients travel to clinics only to discover no slot is available. The goal was to design a simple, accessible booking experience that reduces uncertainty, minimizes wait times, and restores trust between patients and healthcare providers.

Problem	No reliable way to book clinic appointments — patients rely on phone calls that often go unanswered
Goal	Design an appointment booking system that is fast, clear, and works for low-digital-literacy users
Business Objective	Reduce no-shows, reduce front-desk workload, increase patient satisfaction
Scope	web app
Constraints	Low connectivity, mixed literacy levels, older patient demographic, French/Wolof bilingual context

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User Research

Interviews - Observation - Affinity Mapping - Key Insights

Research Goals

- Understand how patients currently book appointments (workarounds, tools, mental models)
- Identify the biggest sources of stress and uncertainty in the process
- Understand the role of front-desk staff - their workload and frustrations
- Map what 'a good booking experience' looks like in this specific cultural and infrastructural context

Research Methods

Method	Why This Method
5 user interviews (patients)	Rich qualitative data - motivations, emotions, mental models
2 interviews (clinic front-desk staff)	Understand the service provider side of the experience
Contextual observation (clinic waiting area)	See real behaviors, not self-reported ones
Google Forms survey (12 additional patients)	Quantitative validation of interview themes

Interview Participants

Profile	Situation	Key Pain
Awa, 34	Mother of 3, books for family	Can't always answer the phone
Mamadou, 58	Chronic condition, visits monthly	Forgets appointments under stress
Fatou, 27	Young professional, books for herself	Hates wasting lunch break at clinic
Ibrahima, 45	Works long hours, misses callback windows	Never reached by clinic callbacks
Rokhaya, 62	Elderly, relies on family to book	Doesn't know how to navigate phone systems

Key Findings

Pain Point 1 - Unanswered Calls Create a Painful Loop

Patients call the clinic. The line is busy or no one answers. They try again later. The clinic tries to call back. The patient misses the callback. No appointment is booked. The patient gives up or just shows up and waits.

Pain Point 2 - No Confirmation = No Peace of Mind

Even when an appointment is 'confirmed' verbally, patients feel uncertain. 'Did they write it down? Will they remember me?' Without a written or digital confirmation, patients arrive anxious — and sometimes discover their appointment was not recorded.

Pain Point 3 - Wasted Trips Are Common

23% of surveyed patients reported arriving at the clinic to find their preferred specialist was unavailable that day. For patients who travel far or take time off work, this is a significant emotional and financial cost.

Pain Point 4 - Scheduling Is a Black Box

Patients have no visibility into available time slots, specialty availability, or wait times. They book blind and hope for the best. Uncertainty is the dominant emotional state.

Affinity Map Summary

Theme: Communication	Theme: Uncertainty	Theme: Accessibility
Phone calls missed by both sides	No visibility on available slots	Elderly patients need help booking
No SMS or WhatsApp confirmation	No confirmation of booking	Rural patients travel far for nothing
Patients unsure who to contact	Fear of showing up for nothing	Low-literacy users struggle
Front desk overwhelmed with calls	Can't check or modify bookings	Limited smartphone familiarity
No digital channel offered	No reminder before appointment	App must work on slow networks

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Define the Problem

Problem Statement - How Might We - Design Principles

Problem Statement

Patients need a faster, clearer, and more reliable way to book clinic appointments because the current phone-based system creates uncertainty, missed connections, and wasted trips — leaving patients feeling unheard and discouraged from seeking care.

How Might We...

How might we help patients in Senegal book clinic appointments easily and confidently, while reducing uncertainty, eliminating wasted trips, and fitting within the real constraints of their digital environment (mobile-first, low connectivity, mixed literacy)?

Design Principles

These principles guided every design decision in this project:

Principle	What It Means in Practice
Clarity over completeness	Show only what the user needs at each step — no information overload
Confirmation always	Every successful action triggers instant visual + SMS confirmation
Accessible by default	Works on low-end Android, 3G, and for first-time smartphone users
Low-friction, one flow	Booking completes in under 3 minutes from start to finish
Culturally appropriate	Language, imagery, and tone reflect Senegalese healthcare context

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User Persona

Based on research - not assumptions

AWA 34 years old Mother of 3 · Dakar <i>Books appointments for the whole family</i>	Background Works part-time, manages a busy household. Books clinic appointments for herself and her three children. Relies on her Android smartphone for everything — messaging, mobile money, social media. Goals ✓ Get quick confirmation — no uncertainty ✓ Avoid wasting a trip to the clinic ✓ Book appointments outside of work hours ✓ Receive reminders so she doesn't forget Frustrations ✗ Phone line always busy or no one answers ✗ Misses the clinic's callback while at work ✗ No written confirmation — she forgets the time ✗ Can't change the appointment without calling again
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Secondary Persona: Oumar, 58 - The Monthly Patient

Oumar manages a chronic condition and visits the clinic monthly. He is not comfortable with smartphones and relies on his daughter to book appointments for him. His key need: the system must be simple enough for a non-digital-native to use or delegate. This persona drove our decision to keep the interface to maximum 3 screens with large tap targets.

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User Journey Map

Current State → Emotional Low Points → Design Opportunities

Current State Journey (Before Design)

This is the journey Awa experiences today — before any redesign. Each step reveals a pain point and a design opportunity.

1. Decides to book	2. Calls the clinic	3. Line busy / no answer	4. Tries again	5. Gets appointment	6. Day before — no reminder	7. Arrives at clinic
Hopeful	Anxious	Frustrated	Still anxious	Relieved but uncertain	Anxious — did she get the right time?	May have to wait 2h+
Trigger: easy entry point	Missed call = lost patient	Hold music / callback option	Live slot availability	Instant digital confirmation	Automatic SMS/WhatsApp reminder	Queue status in real time

Row 1: Journey Step Row 2: Emotional State Row 3: Design Opportunity

Key Insight from the Journey Map

The biggest emotional low points occur at Step 3 (unanswered call) and Step 6 (no reminder). These are the highest-leverage design opportunities: an instant digital booking channel eliminates Step 3, and an automated reminder eliminates Step 6. Solving these two moments transforms the entire experience.

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Ideation

How Might We - User Flows - Key Decisions

Ideation Methods Used

Method	What It Produced
Crazy 8s (8 ideas in 8 minutes)	12 solution concepts — narrowed to 3 viable directions
User flow mapping	Defined the core 3-step booking flow
Paper wireframes	Low-cost visual exploration before prototyping
Team critique session	Eliminated 2 concepts based on feasibility + accessibility

The 3 Concepts Explored

Concept A	Concept B ✓ Chosen	Concept C
Full app	Web app (browser-based)	WhatsApp-only booking bot
Rich features + push notifications	No download required	Ultra-familiar interface
Requires download	Works on any smartphone	Powerful but complex to build
Rejected: adoption barrier for low-tech users	WhatsApp confirmation + SMS fallback	Deferred: excellent v2 feature

Why Concept B Won

A web app requires no download — critical for users with limited storage or data budgets. It works across all devices, loads on 3G, and can send confirmations via WhatsApp or SMS. It balances accessibility with enough functionality to solve the core problems identified in research.

Core User Flow

The booking flow was deliberately kept to 5 steps maximum, with clear progress indicators at each stage:

Step 1 Enter / Select Clinic	Step 2 Choose Specialty	Step 3 Select Date and Time	Step 4 Enter Contact Info	Step 5 Confirmation + WhatsApp/SMS
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Key Design Decision: Confirmation as the Core Feature

After research, we learned that confirmation was the #1 unmet need. Every design decision was made to serve this: instant on-screen confirmation, followed immediately by a WhatsApp message (primary) or SMS (fallback). The confirmation screen was designed as the most important screen in the app — not an afterthought.

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Wireframes

Low-Fidelity → Mid-Fidelity → Key Screen Decisions

Wireframing started on paper — deliberately rough. This sequence matters: low-fidelity first forces focus on structure and flow, not visual polish.

Key Screens Designed

Screen	Design Rationale
Home / Entry Point	Single clear CTA: 'Book an Appointment' — no navigation overload
Specialty Selection	Visual cards with specialty icons — supports low-literacy users
Date and Time Picker	Monthly calendar with availability indicators (green/grey dots)
Patient Information	Minimal fields: Name + Phone only — reduces friction
Booking Confirmation	Full appointment summary + WhatsApp button + SMS option

Low-Fi to High-Fi: The Progression Matters

Why We Start with Paper

Wireframes on paper cost zero — they can be thrown away without loss. Starting rough allows the team to test structural decisions (does this flow make sense?) before investing in visual design. In this project, the paper wireframe session revealed that our original 6-step flow could be collapsed to 4 steps — a discovery that would have been expensive to make after building.

Key UX Details in the Wireframes

- Large tap targets (minimum 48x48px) — all interactive elements designed for elderly users and imprecise touch
- Progress bar on every screen — users always know where they are in the flow
- Back button always visible — reduces anxiety: 'I can go back if I made a mistake'
- Calendar designed mobile-first: week view default, month view on tap — avoids overwhelming small screens
- Specialty icons designed to be understood without reading — serves low-literacy users

High-Fidelity Prototype

Style	Clean, calm, high-contrast — healthcare context requires trust, not excitement
Color System	Primary teal (#0A7373) for actions; warm white backgrounds; error states in amber
Typography	Large body text (18px min) — accessibility for older users; high contrast ratios (WCAG AA)
Language	French primary; Wolof key terms in parentheses where relevant

Design Decisions — Justified by Research

Design Decision	Research Justification
Large primary button: 'Book Appointment'	Users said they struggled to find the right action — large CTA eliminates ambiguity
Specialty shown as icon + label	Icon-only failed low-literacy test; text-only missed non-readers; combined solved both
Calendar shows availability dots	Users asked 'how do I know if there's space?' — visual availability reduces calls to clinic
Confirmation screen shows full summary	Users feared the clinic 'would not remember' — full written summary = trust
WhatsApp button above SMS option	80%+ of users prefer WhatsApp; SMS is the accessibility fallback for feature phones

Design Principle in Action: Confirmation Screen

In most apps, the confirmation screen is treated as the end — a receipt. In this design, confirmation is the centerpiece. It shows: doctor name, date, time, clinic address, and a large 'Save to WhatsApp' button that sends all details to the patient's WhatsApp immediately. Design reasoning: 'I used an immediate WhatsApp delivery because our research showed that patients lose paper receipts and forget verbal confirmations. A WhatsApp message is always searchable and always with them.'

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Usability Testing

5 Users - 2 Tasks - Real Observations

Participants	5 users matching research personas (3 patients, 1 caregiver, 1 elderly)
Format	Moderated remote sessions (30 min each) via phone + screen share
Tasks Given	Task 1: Book an appointment in cardiology for next Tuesday at 10 AM
	Task 2: Your appointment changed — reschedule to Thursday afternoon
Success Metric	Task completion rate · Time on task · Observed hesitations · Satisfaction score (1-5)

Results

4/5 Completed Task 1 without help	3/5 Completed Task 2 independently	3.8/5 Average satisfaction score
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Issues Found

Issue Observed	Severity
Calendar was confusing on first visit	High — 3 of 5 users hesitated
'Reschedule' option not found easily	High — caused Task 2 failures
Confirmation screen: 'WhatsApp' button not immediately understood by oldest user	Medium — resolved after 10 seconds
'Back' button missed during rescheduling	Low — minor friction only

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Iteration

What We Changed After Testing — and Why

Before Testing	After Iteration
Complex monthly calendar as default view	Week view default with 'See full month' toggle
Reschedule buried in confirmation email	Reschedule button prominent on confirmation screen
WhatsApp button labeled 'Send to WhatsApp'	Relabeled: 'Save appointment on WhatsApp' — clearer intent
Booking flow: 5 steps	Booking flow: 4 steps (combined name + phone fields)
No visual indicator of calendar availability	Green/grey dots on calendar show real-time availability

Key Iteration: Calendar Simplification

Based on usability testing, I simplified the booking flow from 5 steps to 4 by merging the patient information fields. I also replaced the default monthly calendar (which overwhelmed users with too many choices) with a weekly view showing only the next 7 days — dramatically reducing decision fatigue. 'See full month' is available on tap for users who need to book further ahead.

Results and Impact

Measured Outcomes from Post-Iteration Testing

40% Faster task completion after redesign	5/5 Users completed Task 1 after iteration	4.4/5 Satisfaction score (up from 3.8)
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Before Redesign	After Redesign
Phone-only booking — unanswered calls	Digital booking in 4 steps, under 3 minutes
No confirmation or reminder	Instant WhatsApp confirmation + reminder 24h before
5-step flow with confusing calendar	Week-view calendar with availability dots
Patients arrive uncertain and stressed	Patients arrive informed and confident
Front desk manages all bookings manually	Front desk focuses on care, not coordination

Projected Business Impact Based on usability results and industry benchmarks for comparable clinic digitization projects: • Estimated 30–40% reduction in no-shows (reminder = accountability) • Estimated 50% reduction in front-desk phone workload • Projected 25% increase in patient satisfaction scores • Accessibility improvement: elderly and low-literacy users can now book independently
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Reflection

What I Learned · What I Would Do Differently · Next Steps

Learning 1 — Simplifying decisions is the designer's highest-value work

This project taught me that the most important UX skill is not making things beautiful — it is making choices disappear. Every time I reduced a decision the user had to make (fewer calendar options, merged form fields, pre-written WhatsApp message), task completion improved and satisfaction scores rose. Simplicity is not a visual style; it is a design outcome.

Learning 2 — Research reveals solutions that logic alone never would

Without talking to real users, I would have built a complex feature-rich app. Research revealed that what patients actually needed was simple: confirmation and a reminder. The solution that came from research was smaller, cheaper to build, and far more effective. The lesson: the best design brief comes from listening, not from assuming.

Learning 3 — Context is a design constraint, not a limitation

Designing for low-connectivity, mixed-literacy, mobile-first users in Senegal forced me to make better decisions: lighter pages, larger tap targets, icon-plus-label components, WhatsApp over email. These constraints didn't limit the design — they focused it. A design that works for the most constrained user works for everyone.

What I Would Do Differently

- Involve a clinic staff member as a co-designer from the start — the back-end scheduling system matters as much as the patient-facing app
- Test with at least 2 elderly users in the first round — their needs revealed important issues that a younger sample would have missed
- Prototype the WhatsApp confirmation flow earlier — this was the most critical feature but was only fully tested late in the process

Next Steps (V2 Roadmap)

Feature	Why It Matters
WhatsApp booking chatbot	Removes web entirely for WhatsApp-native users
Doctor profile pages	Helps patients choose specialist with confidence
Real-time queue status on day of visit	Eliminates last-mile uncertainty

Patient medical history (with consent)	Enables better care continuity
Multilingual: French + Wolof	Full accessibility for non-French speakers
